Examiner only

13. (a) Two students were discussing how to prepare chloroethane. One student suggested reacting chlorine with ethane but the other student said that the yield from this reaction was poor and that it was better to react hydrogen chloride with ethene.

(i) State the **type** of reaction involved in each case. [2]

ethane and chlorine

....

ethene and hydrogen chloride

....

(ii) Suggest why the reaction between ethene and hydrogen chloride gives the better yield. [1]

....

....

(iii) Give the mechanism for the reaction between ethene and hydrogen chloride. [4]

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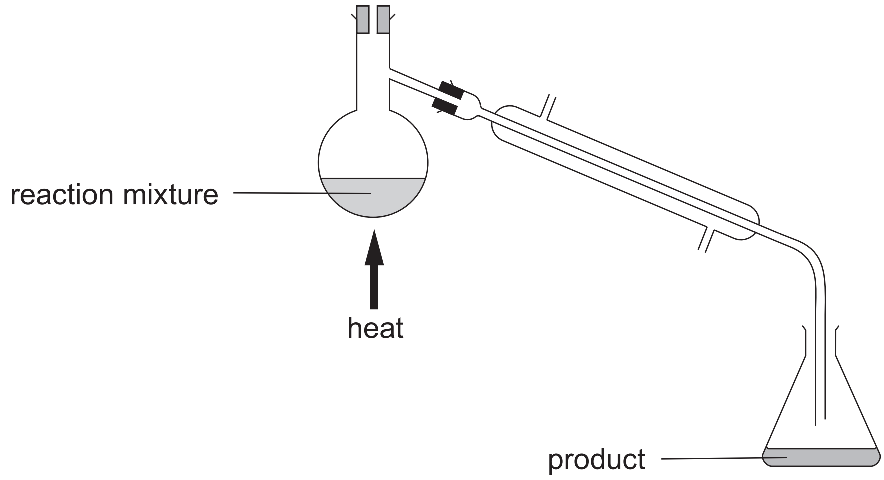
(b) The equation for the reaction between ethanoic acid and methanol to form an ester is shown.

$$CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O$$

(i) In the preparation of this ester, 25 g of ethanoic acid reacted with excess methanol. If the percentage yield of the reaction was 34 %, calculate the mass of ester formed. [3]

Mass = ... g

(ii) The diagram shows the method used to separate the ester from the reaction mixture.



Draw on the diagram

- the position of the thermometer
- the direction of water flowing through the condenser [2]

Turn over.

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(iii) Addition of concentrated sulfuric acid increases the yield of ester. Explain this observation. [2]

....

....

....

(c) Draw the structure of the ester formed when propan-2-ol reacts with methanoic acid. [1]

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END OF PAPER